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Sub ject : DSA

Unit 3, S11, SLO2 - Circular Queue - Enqueue Operation

Algorithm to insert an element in a circular queue
Step 1: IF FRONT = ____ and Rear = ____________
Write OVERFLOW
Goto step 4
[END OF IF]
Step 2: IF FRONT=-1 and REAR=-1
SET FRONT = REAR = _______
ELSE IF REAR = ____________ and FRONT != _______
SET REAR = 0
ELSE
SET REAR = __________
[END OF IF]
Step 3: SET QUEUE[REAR] = _________
Step 4: EXIT

Answer: Step 1: IF (FRONT = 0 and REAR = MAX - 1) OR (FRONT = REAR + 1)
Step 2: IF FRONT = -1 and REAR = -1
SET FRONT = REAR = 0
ELSE IF REAR = MAX - 1 and FRONT != 0
SET REAR = 0
ELSE
SET REAR = REAR + 1
Step 3: SET QUEUE[REAR] = NUM
Step 4: EXIT

Fill in the Missing Programming Code for Circular Queue insert():

Answer: void insert() {
```
int num;
printf("\nEnter the number to be inserted in the queue : ");
scanf("%d", &num);
if ((front == 0 && rear == MAX - 1) || (front == rear + 1))
printf("\n OVERFLOW");
else if (front == -1 && rear == -1) {
front = rear = 0;
queue[rear] = num;
}
else if (rear == MAX - 1 && front != 0) {
rear = 0;
queue[rear] = num;
}
else {
rear = rear + 1;
queue[rear] = num;
}
}
```